Adithya Rao Kalathur

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LinkedIn | GitHub | Portfolio | Certificates

Profile

ML engineer focused on **agentic AI systems, developer tools, retrieval architectures, and production ML deployment**. Built RAG pipelines, deep learning models for medical imaging, and full-stack applications. Proficient in **Python, PyTorch, FastAPI, and Docker**. Seeking **ML engineering or AI backend roles**.

Education

Manipal School of Information Sciences (MSIS) — Manipal, India
Master of Engineering in Computer Science and Engineering; CGPA: 7.96/10 — *2025 – 2027 (Expected)*

Manipal Institute of Technology (MIT) — Manipal, India
Bachelor of Technology in Computer and Communication Engineering; CGPA: 7.06/10 — *2020 – 2024*

Technical Skills

Languages: Python, SQL, Java, C, C++
ML & Deep Learning: PyTorch, Scikit-Learn, TensorFlow, U-Net, Transfer Learning, Grad-CAM, Hugging Face, XGBoost, BiLSTM
RAG & LLM Systems: LangChain, LangGraph, FAISS, Sentence Transformers, BM25, Cross-Encoder Reranking, Ollama, RAG Pipelines
NLP & Data Science: Natural Language Processing, Text Classification, Pandas, NumPy
Backend & Web: FastAPI, React.js, Node.js, PostgreSQL, REST APIs, Streamlit
Data & Analytics: Power BI, DAX, ETL Pipelines, SQL
Systems & Tools: Linux, Shell Scripting, Git/GitHub, Docker, Kubernetes, OpenCV, API Integration

Work Experience

Full Stack Web Development Intern — Jun 2023 – Jul 2023
Thaniya Technologies [Certificate] — *Mangalore, India*

- Built and deployed a full-stack movie review and recommendation platform using **React.js, Node.js, and PostgreSQL**; shipped to production on Render via cloud hosting [GitHub].
- Designed the relational database schema, implemented REST APIs for all CRUD operations, and integrated backend services with the React frontend.
- Sole developer responsible for UI design, database schema, REST API development, content-based recommendation logic, and cloud deployment on Render.

Freelance Web Developer — Dec 2024 – Mar 2025
Startup Client Project — *Remote*

- Built and deployed a responsive portfolio website for a landscape design startup; managed client communication, iterated on designs based on feedback, and delivered within timeline.
- Structured service and contact pages for desktop and mobile; sole point of contact for requirements, architecture, and deployment decisions.

Projects

CodeIntel-RAG — Agentic Codebase Intelligence Platform [GitHub] — 2025
RAG, Hybrid Retrieval, FAISS, BM25, LangGraph, FastAPI, Ollama, Security Analysis

- Built a self-hosted agentic RAG platform for natural-language querying over Python, Java, JavaScript, TypeScript, and C++ codebases; achieved perfect **Hit@5 (1.00)** and **MRR of 0.845** on a 25-query benchmark against pallets/flask — 8.7% above the semantic baseline.
- Designed a custom 5-signal hybrid retrieval pipeline combining BGE embeddings + FAISS, BM25Okapi, structural call-graph traversal, file-metadata filtering, and name-token concept matching.

- Integrated a LangGraph ReAct agent with five autonomous tools including a Bandit + regex security scanner (CWE/OWASP Top 10) and multi-hop dependency analyser, enabling multi-tool reasoning chains in a single query turn.
- Built a production-style FastAPI backend with 15+ endpoints, streaming SSE, caching, and containerised deployment via Docker Compose; frontend includes voice input (Whisper) and interactive Pyvis call graphs.

Deep Learning-Based SCC Grading from Histopathology Images [GitHub] 2025

Deep Learning, Transfer Learning, Computer Vision, Explainable AI (XAI), Grad-CAM, Streamlit, PyTorch

- Benchmarked 5 transfer-learning CNN backbones (InceptionV3, EfficientNetB0, DenseNet121, MobileNetV2, ConvNeXt-Tiny); best model achieved **96.5% test accuracy** and **macro-F1 0.94** on a held-out dataset.
- Deployed an interactive Streamlit web app with real-time Grad-CAM visualisations and LLM-generated prediction summaries, making model decisions interpretable for clinical users.
- Addressed class imbalance through class-weighted loss and extensive augmentation; applied early stopping, learning rate scheduling, and model checkpointing to improve generalisation and reduce overfitting.

Night Vision Enhancement using Deep Learning [GitHub] 2025

U-Net, ResNet34, PyTorch, OpenCV, Computer Vision, Image-to-Image Translation

- Built two low-light image enhancement pipelines — a vanilla U-Net and a ResNet34 encoder-decoder U-Net — trained separately on the LOL dataset (485 pairs) and the ImageNet Night Vision dataset (40K+ pairs) using paired image-to-image translation.
- Achieved **PSNR of 18.84 dB** and **SSIM of 0.71** on the ImageNet Night Vision test set (5000 images) with the pretrained ResNet34 U-Net trained using L1 loss and ReduceLROnPlateau scheduling.
- Designed training pipelines with combined MSE + SSIM loss, ColorJitter augmentation, and residual learning; evaluated outputs quantitatively using PSNR and SSIM across multiple model variants.

FMCG Supply Chain Analytics Dashboard [GitHub] 2025

Power BI, DAX, SQL, ETL, Data Analytics

- Built an end-to-end Power BI dashboard tracking supply chain KPIs — OTIF, LIFR, and VOFR — across 3 cities and 50+ product categories for Atliq Mart.
- Developed 20+ DAX measures and SQL-based ETL pipelines for root-cause analysis; identified a **15% fulfillment service gap** across key accounts that directly informed operational strategy.

Publications & Research

SQL Injection Detection: Comparative Study of SVM, RF and Hybrid Models [Publication] Oct 2025

SVM, Random Forest, BiLSTM, NLP, Hybrid Ensemble, Cybersecurity

- **First Author** — Published as a book chapter in *Connecting Intelligence: Trends in Computation and Data Communication*, Taylor & Francis / CRC Press (ISBN: 9781003773504); presented at ICCIDC 2025, Bali, Indonesia [Certificate]
- Achieved **97.84% accuracy**, **98.09% precision**, and **97.57% recall** with a hybrid SVM-RF ensemble on 30,920 labeled SQL queries; a BiLSTM model further reached **98.75% accuracy**, validating sequence-based NLP for automated threat detection.

Certifications

- IBM Data Science Professional Certificate — Coursera, 2024
- Google Data Analytics Professional Certificate — Coursera, 2024
- Generative AI with Large Language Models — DeepLearning.AI, 2025
- Google Cybersecurity Professional Certificate — Coursera, 2024

Languages

English: Professional proficiency
Kannada: Native
Tulu: Native
Hindi: Conversational